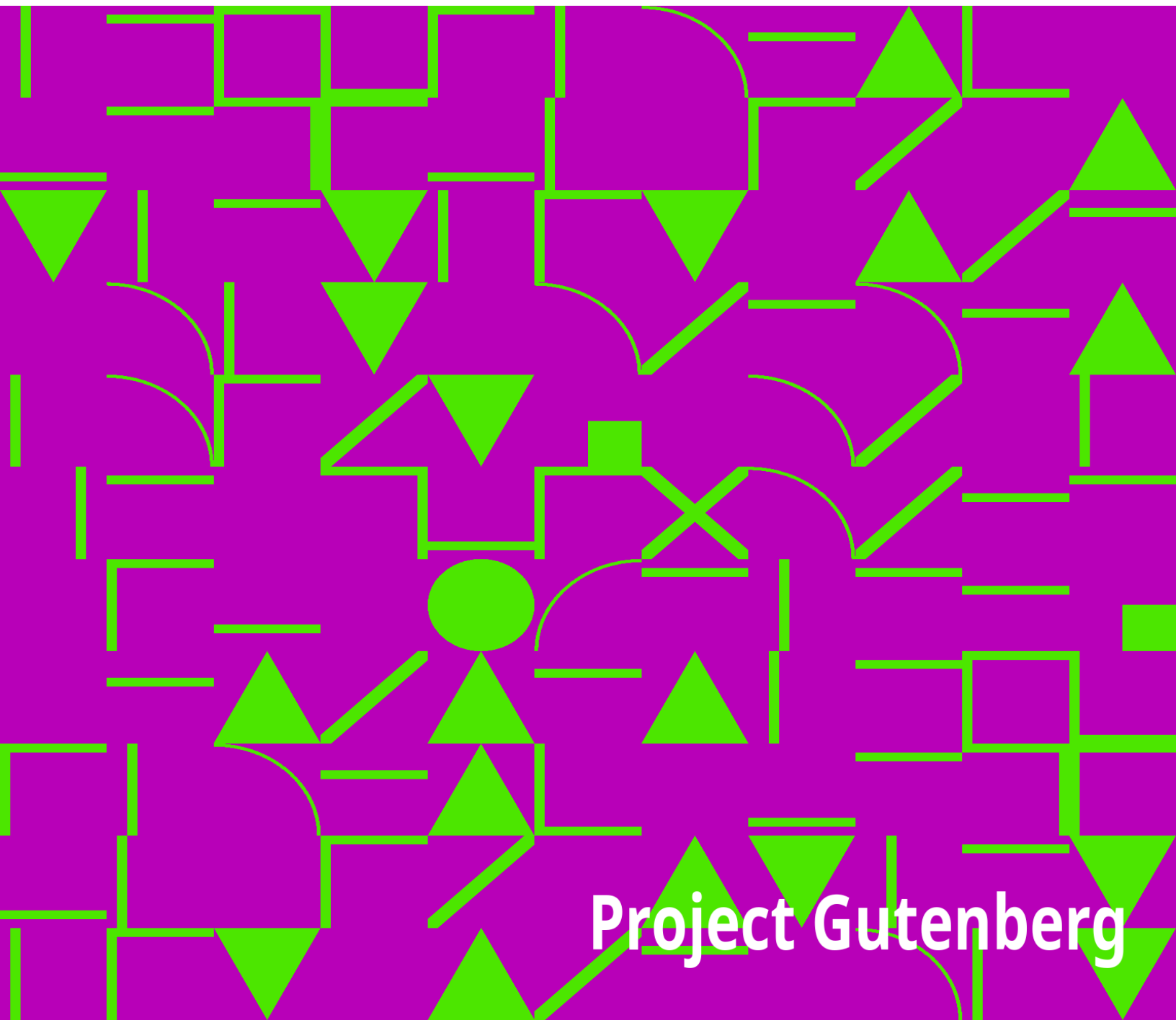


# **Geological Report on Asbestos and its Indications, in the Province of Quebec, Canada**

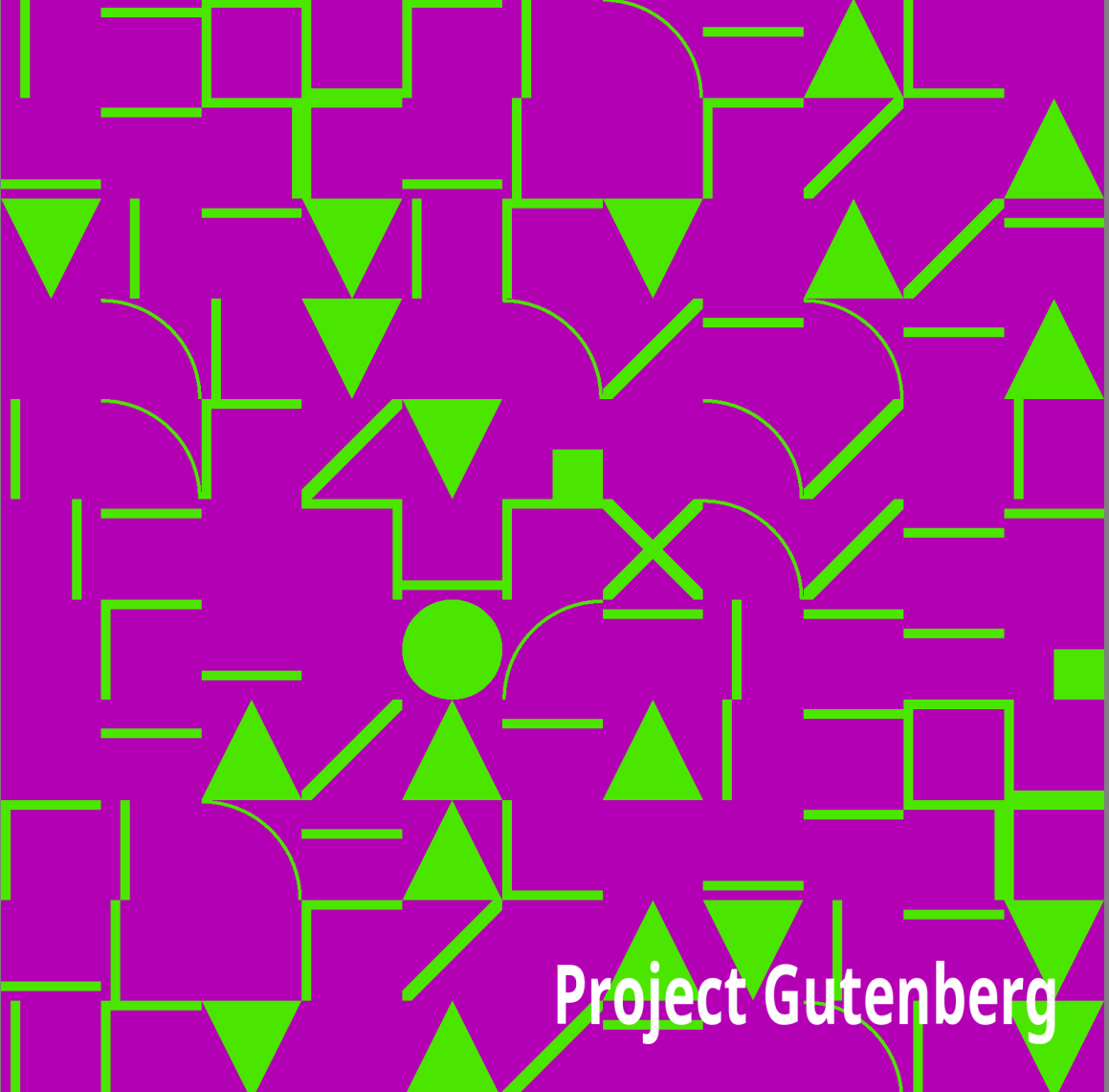
**Lucius J. Boyd**



**Project Gutenberg**

# Geological Report on Asbestos and its Indications, in the Province of Quebec, Canada

Lucius J. Boyd



Project Gutenberg

## **The Project Gutenberg eBook of Geological Report on Asbestos and its Indications, in the Province of Quebec, Canada**

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Geological Report on Asbestos and its Indications, in the Province of Quebec, Canada

Author: Lucius J. Boyd

Release date: September 4, 2010 [eBook #33630]

Most recently updated: January 6, 2021

Language: English

Other information and formats: [www.gutenberg.org/ebooks/33630](http://www.gutenberg.org/ebooks/33630)

Credits: Produced by The Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by the Canadian Institute for Historical Microreproductions ([www.canadiana.org](http://www.canadiana.org)))

**\*\*\* START OF THE PROJECT GUTENBERG EBOOK GEOLOGICAL  
REPORT ON ASBESTOS AND ITS INDICATIONS, IN THE PROVINCE  
OF QUEBEC, CANADA \*\*\***

---

# **GEOLOGICAL REPORT ON ASBESTOS**

---

GEOLOGICAL REPORT

ON

ASBESTOS,

AND ITS INDICATIONS, IN

THE PROVINCE OF QUEBEC,

CANADA.

---

LONDON:  
E. FORSTER GROOM, 15, CHARING CROSS, S.W.

---

1889.

*[All rights reserved.]*

---

# GEOLOGICAL REPORT

## ON

### ASBESTOS,

## AND

### ITS INDICATIONS IN THE PROVINCE

## OF

### QUEBEC, CANADA.

---

HAVING been called upon to make a close and careful examination of the geological formations in the eastern townships of Garthby, Wolfstown, and Coleraine, situated in the province of Quebec, Canada, I gave special attention to the distribution of the Asbestos-bearing rocks (serpentine), which have been, in my opinion, heretofore only partially traced. Perhaps this was owing to the difficulties which had to be encountered from the thick undergrowth which in many places rendered it almost impossible to penetrate sufficiently in order to make a *true* report as regards the "existence," "location," and "association" of these rocks.

Admirable reports have been written by R. E. Ellis, LL.D., Dr. Hunt, and others, on the origin and distribution of the serpentines, and have been fully discussed and ventilated. Still, though various opinions have been expressed

upon the subject, they appear to differ in many respects. I mention these facts as possibly one inexperienced or unacquainted with the country might consider it strange that a thorough examination of the Asbestos properties had not been followed. Yet the causes I have mentioned above, as well as the difficulties I had to contend with during the months of heavy snowfall, lead me to believe that my *confrères* (geologists) were disinclined to follow up a correct and actual prospectus of these valuable serpentinous localities.

Before locating, or going into details of these classes of rock as a mineral repository, I intend to treat on the subject as regards their mode of existence and origin.

Serpentine is diffused under the head of "metamorphic rocks," while, in the widest sense, according to Studor and others, mineral metamorphism means every change of aggregation, structure, or chemical condition, which rocks have undergone subsequently to their deposition and stratification, or the effects which have been produced by forces other than gravity and cohesion.

There fall under this definition the discolouration of the surface of, for instance, black limestone, by the loss of its carbon, the formation of brownish red crusts in rocks of limestone, sandstone, many slatestones, shales, granite, &c., by the decomposition of compounds of iron, finely disseminated in the mass of the rock, the change in rocks consequent in the absorption of water, and the crumbling of many granites and porphyries into gravel, occasioned by the decomposition of the mica and felspar.

In its more limited sense the term "metamorphic" is confined to those changes of rock which are produced directly or indirectly by agencies seated in the interior of the earth. In many cases the mode of change may be explained by our physical or chemical theories, and may be viewed as the effects of temperature or of electro-chemical actions adjoining rocks or connecting communications with the interior of the earth, also distinctly point out the seat from which this change proceeds. In many other cases the metamorphic process itself remains a mystery, and from the nature of the products alone do we conclude that such a metamorphic action has taken place.

Serpentine is generally believed to have been originally deposited as a sediment, and to have acquired its present compact crystalline character through the subsequent action of various chemical, or mechanical, agencies. It is known to be a *hydrated silicate of magnesia* with about equal parts of silica and magnesia, and contains 12 per cent. of water with varying proportions of iron, chromium manganese, alumina and lime, has a specific gravity of 2.7, and weighs about 169 lbs. to the cubic foot. It is found both in a soft and very compact state, of a waxy lustre, with many different shades of beautiful green which give it a mottled appearance like a serpent, hence the origin of its name "serpentine," or ophite. It is called "ranocchia" by the Italians, from the appearance it bears to the "frog," and, on account of its susceptibility to a high polish, is greatly valued as a marble for interior ornamental purposes, more than exterior, as it weathers rapidly. In Galway, Ireland, it is found in large quantities, and called "serpentinous marble," or "ophi-calcite." It is also to be found in other parts of the world, as in the Pyrenees, Alps of Dauphing, Mount St. Gothard, Italy, Sweden, Ural Mountains, Silesia, New South Wales, Savoy, Corsica, Cornwall, Scotland, and other places too numerous to mention; but in Canada the finest and most crystalline serpentine is to be found forming great belts of over 100 miles long and several thousand feet in breadth. There it associates with the dioretic, or volcanic, rocks, and is, according to Dr. Ells, without any doubt, "An alteration product of a dioretic rock rich in olivine." It is sometimes very difficult to distinguish the mineral constituents in many of the metamorphic rocks, but diorite is always considered to be composed chiefly of felspar and hornblende, which composition enters largely into the serpentines. Actinolite, tremolite, &c., and many other minerals, are sometimes found associating with it.

There are many valuable properties attributed to serpentine, and I am of opinion that the time is not far distant when it will be commercially considered an invaluable substance, and this on account of its refractory properties. I may also mention that it can be extensively used in the manufacture of crucibles, &c. Its soft and unctuous qualities (especially where it is found associated with "steatite," or "soapstone," which is often to be seen in large quantities) renders it easy to be worked, and, if reduced to a powder, could be moulded in bricks which the most intense heat will not affect. One of the chief properties it contains, and one which the



serpentine of Lower Canada is so famous for, is the Asbestos, crysotile, or fibrous serpentine. This valuable and important mineral product is found in paying quantities only at certain points in the extensive serpentine reefs, and was first mined as an article of commerce in Canada in 1878, and has now become a regular and rapidly-developing industry.

On account of its incombustible and indestructible qualities, is extensively used in steam, hydraulic, and electrical machinery. It has been adopted by the Admiralty for engine packing, in Her Majesty's war-ships. It is spun into six-fold yarn, with a tensile strength of 40 lbs. and upwards, is manufactured into cloth, as clothing for firemen, and covering hose-pipes, in fire brigades, and also engine purposes, as well as drop-curtains, and general stage scenery, and is employed by the principal railway and steamship companies, collieries, ironworks and all classes of factories, and, in the manufacture of the new Asbestos grates and stoves, is finding for itself a large market. Messrs. Bell & Company of London, who are the largest Asbestos users in the world, have adopted it in the manufacture of over 50 special purposes in connection with steam engines and general machinery, and, as a lubricant, Asbestoline ranks in the first degree.

There have been many mines started in Canada by people of the farming class, as well as by companies, and "cotton," as Asbestos is locally termed, has been found in large quantities within a few feet from the surface, in veins from 1/4 to 6 inches and more in length of fibre. In Italy, Asbestos is found, measuring up to 6 feet, in fibre, and chemically speaking, there is no difference between it and Canadian, except that the latter, though shorter in fibre, is much more compact and crystalline, and purer in every sense of the word than can be obtained in Italy, so much so, as I understand, that users of Italian material have virtually abandoned it for Canadian. Although I have no doubt but that Italian Asbestos has its own special purposes.

The greatest depth reached in Canada is 130 feet in open workings. No timbering or extensive machinery is used in the manipulation of the mines, as the "cuts"—being usually in the mountain side—afford a natural drainage, and dumpage.

Having blasted the rock, the first process of extraction is termed "cobbing," which means breaking off the adhering serpentine from the

Asbestos vein, this being manual work done by boys. The fibre is packed in sacks, each weighing 100 lbs., and in some cases 200 lbs. are shipped by the local railroad company to Montreal or New York at something about 10 cents and 20 cents per sack.

Asbestos is sorted into three qualities, and priced thus:—

|     |          |         |    |      |      |     |        |     |     |    |       |      |
|-----|----------|---------|----|------|------|-----|--------|-----|-----|----|-------|------|
| 1st | quality, | selling | at | mine | \$80 | to  | \$200, | per | ton | of | 2,000 | lbs. |
| 2nd | "        | "       | "  | 60   | "    | 70, | "      | "   | "   |    |       |      |
| 3rd | "        | "       | "  | 25   | "    | 50, | "      | "   | "   |    |       |      |

Some inferior quality, at a very low price, is used by the Asbestos Mining and Manufacturing Company of Quebec.

The workmen are principally French Canadians belonging to the neighbouring villages, and the wages paid them are—

Miners (without board), \$0.90 (3/9) to \$1.25 (5/2), per day.

Pickers and cobbers, \$0.40 (1/8) " \$0.70 (2/11), "

The cost of extraction is taken from \$20 to \$25 per ton; this includes local administrations and all other expenses connected with the mine, and with the adoption of machinery and the use of air-compressed drills the cost of actual mining will be reduced to at least 30 per cent.; so taking an average price of about \$70 per ton, a net profit of from £8 to £9, or \$45, is obtainable per ton of raw material.

In 1886 the total amount of Asbestos, taken from all the mines, may be estimated at 1,500 tons, and of the amount returned last year (1888), all but 400 tons were from the Quebec province mines, and of these Thetford turned out 2,560 tons, and Black Lake 950, or together three-fourths of the whole out-put. The 400 tons were from Bridgewater, in the province of Ontario, a somewhat different class of mineral, which is generally used in the manufacture of fire-proof roofing.

As regards the indications of Asbestos, it is a general recognised fact, and one that may be depended on, that not alone in Canada, but indeed all other places where Asbestos-bearing serpentine is found, the existence of

Asbestos, or "Amianthus," is noticed when the serpentine is exposed, and presents a rusted, sometimes greyish and broken appearance, due to decomposition or weathering, or covered with a thin layer of soil. Small veins of Asbestos are to be seen forming a network on the surface of the rock. If closely examined there may be noticed the indications of a fault which, in the eastern townships of Quebec, has generally a direction of N. 40° E., this fault appearing in all openings where a good show of mineral is to be seen, presenting a wall either in a vertical position or at an angle, which is preferred to be not greater than 30°. From this wall, at a varying distance of from 5 to 20 feet, will be found another, sometimes parallel to, or at an opposite angle; in this latter case, if these walls be worked down, they will be found to either meet, forming a trough-like appearance, or to change their course in a downward direction, leaving only a few feet from each other at the narrowest point, and then diverge to an unlimited depth. In this case their faces will have a slicken appearance, smeared over with thin layers of imperfect Asbestos, or crysotile, now and then compact, fibrous hornblende, up to 24 inches in length, of various colours, and rich deposits of olivine, in rare cases small quantities of "ground ivory" with many other admixtures.

The condition of the serpentine within these walls is greatly distorted, containing many small veins of Asbestos varying from mere threads to 2 and 3 inches in thickness, and sometimes deposits of grains of magnetic iron or magnetite with traces of chromic iron, which in some localities break the continuity of the fibre, veins of rich white crystalline matter (perhaps calcareous) with large deposits of "soapstone," or steatite, associated with "serpentite." Such contorted out-crops are indications of rich veins of Asbestos, which will be found to both increase in quantity and quality the deeper they are worked. And in the case where the walls are parallel and the filling matter in the same contorted condition, it is inevitable, in order to obtain a good fibre, considerable depth should be reached.

The serpentine, which constitute these walls, will also be found to proportionally become more compact, and less associated with impurities, and contain the finest quality and lustre of fibre.

A very interesting phenomenon may be noticed at some of the mines in connection with this contorted matter. It is the transposition of the serpentine into Asbestos fibre, by the action of the atmosphere. This is to be seen on the dumps where the filling matter and cobbled rock is exposed. In one or two cases I have seen large quantities of broken rock changed into fibre after a few years, by atmospherical chemical agencies.

In so many cases I find people are prejudiced from going deeper than a few feet from the surface, as not finding a copious supply of Asbestos there, when *good* indications are shown they become disheartened.

Therefore, from these practical facts it will be seen that in order to get the best results it is necessary to work at the lowest possible level when a favourable out-crop is shown, as, possibly, working at a high elevation on the out-crop may be a mistake, where a lower point is available.

There are good indications of Asbestos where the serpentine is crossed by quartzose, gneiss, or "traverse dykes," and some valuable finds have been made at the junction with the dioretic rocks.

When the serpentine is found dark in colour, to have a granular appearance, containing many dark grains of, perhaps, felspathic crystals, the Asbestos will be of a dark, dull, translucent lustre, very compact, and easily fluffed to a fine silken fibre. The admixtures of hard and soft serpentine, where not effected by a fault, may sometimes be regarded as a doubtful indication of an immediate find, but if its hardness increases on descending, and colour becomes more uniform, from a light emerald green with a whitish admixture, to a dark olive, and containing numerous small veins of fibre, the conclusions may be considered as favourable to rich deposits of Asbestos.

In conclusion I may add that the foregoing remarks, as regards the indications of this valuable mineral, are based on my personal geological experience, and the reliable information of the managers of the various Asbestos mines in Canada, whose opinions have greatly aided me in my recent prospection, and I trust that this pamphlet will not alone be a benefit to them, but to the Asbestos industry, which I feel assured will be one of the most prominent in the province of Quebec.

LUCIUS J. BOYD,  
C.E., F.R.G.S.I.

\*\*\* END OF THE PROJECT GUTENBERG EBOOK GEOLOGICAL  
REPORT ON ASBESTOS AND ITS INDICATIONS, IN THE PROVINCE  
OF QUEBEC, CANADA \*\*\*

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

# THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at [www.gutenberg.org/license](http://www.gutenberg.org/license).

## **Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works**

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to



check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website ([www.gutenberg.org](http://www.gutenberg.org)), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format

must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you

received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you ‘AS-IS’, WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

## **Section 2. Information about the Mission of Project Gutenberg**

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at [www.gutenberg.org](http://www.gutenberg.org).

### **Section 3. Information about the Project Gutenberg Literary Archive Foundation**

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at [www.gutenberg.org/contact](http://www.gutenberg.org/contact)

### **Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation**

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the

number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

## **Section 5. General Information About Project Gutenberg electronic works**

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:  
[www.gutenberg.org](http://www.gutenberg.org).

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.